

Preparing OCT blocks of kidney tissue

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1. Dissect out kidneys rapidly and store briefly (up to 20 min or so) in ice cold PBS.
2. Process through OCT only as many kidneys as you plan to freeze in one block at a time. The OCT is hyperosmotic and tends to suck the water out of the tissue it touches, ruining the edges of the kidney, where a lot of interesting stuff is happening. So faster is better. But, at the same time, you need to totally rinse the PBS away, because it can interfere with the sectioning.
3. Place kidneys in OCT in a 60 mm plate cooled with ice. Use OCT that has been pre-cooled on ice for at least ten minutes.
4. Quickly mix kidneys in with OCT, then transfer carefully to a new 60 mm plate with cold OCT, quickly mix again, and place in a tinfoil mold with cold OCT covering the bottom.
5. Cover kidneys with additional OCT and position kidneys near each other in mold, in central position, not too near the top.
6. Immediately freeze in 2-methylbutane isopentane that is in a pyrex beaker resting in liquid nitrogen. The isopentane should be frozen. Hold the tinfoil mold with forceps to keep vertical and gently move to keep from freezing in the isopentane and to improve thermal contact.
7. When the OCT is completely frozen place the mold in dry ice, and then store long term in liquid nitrogen freezer.